



Skills/Careers/Resume/ Networking *Engineering Physics*

Office of Career Development &
Professional Engagement

Skills and Attributes



Physics/Engineering = ?

- Analytical – how things work and don't work
- Problem Solving – logical, structured thinking
- Project Management – research projects
- Systems Thinking – big picture, small details
- Time Management – recognized as challenging curriculum
- Others?

Qualities/Attributes

- Curiosity
- Lifelong Learning
- Learning how to learn
- Innovation
- Entrepreneurship thinking

Career Paths



Career Paths

- *Engineering-Specific*
 - Mechanical, Electrical, Civil, Computer, Biomedical, Industrial, AI/ML

Career Paths

- *Research*

- National Labs, Universities, other Government Organizations (NASA, Naval Research Lab, Defense Advanced Research Projects Agency), some industry (Intel, Google, Microsoft)

Career Paths

- *Aerospace/Aeronautical*
 - NASA, SpaceX, Blue Origin, Orbital ATK, Bigelow, Aerospace Corp., Boeing, Airbus

Career Paths

- *Health Care*

- Medical Physicist – radiation treatment for cancer
 - Hampton University Proton Therapy Institute
- Instrumentation – Olympus Medical, Siemens, GE Healthcare

Career Paths

- *Data Science, Data Analytics, Data Architecture*
 - Digital Marketing - Merkle, Red Ventures
 - Healthcare Informatics
 - Biomedical, Genomics

Career Paths

- *Marine Science*

- NOAA, Army Corps of Engineers, Chesapeake Research Consortium
- Jack Slater '22 – Marine Scientist, VIMS

Career Paths

- *Teaching/Education/Outreach*
 - Higher Education – university, college, community college
 - High School – high demand, alternative certification programs (Noyce Scholarship Program)
 - Museums – Virginia Air & Space Center, Kennedy Space Center

Career Paths

- *Consulting*

- Boston Consulting Group (BCG), Deloitte, Booz Allen Hamilton, McKinsey & Company, IBM, Censeo

Career Paths

- *Science Writing, Communications*
 - National Labs, Publications (Physics Today, Physics World, Science), Websites, National Science Foundation

Career Paths

- *Finance and Actuarial*

- Predictive Modeling -Stevens Capital Management, D.E. Shaw Group, Freddie Mac
- Actuarial – Markel, GEICO

Career Paths

- *Patent Research, Patent Law*
 - U.S. Patent and Trademark Office, CPA Global, Cardinal IP

Career Paths

- *Science Policy*

- The American Association for the Advancement of Science (AAAS), Office of Science and Technology Policy (The White House)

Resumes



What is a Resume?

- Represents you to people who don't know you
- Demonstrates your relevant skills, experiences, and qualifications
- Used to apply to internships, fellowships, jobs, graduate and professional school

Resume Perspective

- Put yourself in the shoes of the reader
- First representation of yourself
- “Active” document
- A resume is
 - Not used to get a job or internship
 - Used to get an interview

What Goes on Your Resume?

- A variety of experiences!
- Think “transferable skills and qualities”
- Think “talking points”
- Think from the employer’s perspective

What Goes on Your Resume?

- Relevant coursework and associated projects
- Research
- Internships
- Memberships and Leadership
- Volunteer
- Jobs – part-time, seasonal

What Goes on Your Resume?

- What makes you different?
- What makes you stand out?
- What could be a talking point beyond the expected content?
- Think talking points

Think in Sections

- Resumes are scanned quickly
- Make it easy for the reader to find information
- Bolding and ALL CAPS

Think in Sections

- EDUCATION
- RELEVANT COURSEWORK
- RELEVANT EXPERIENCE
- RESEARCH EXPERIENCE
- WORK EXPERIENCE
- VOLUNTEER/SERVICE
- MEMBERSHIPS/LEADERSHIP
- SKILLS

Format

Reverse Chronological

- List your most recent experience first then work backwards
- Each section is a timeline
- Place sections according to relevancy

Make it Easy to Read

- Bullets
- Bolding
- Italics
- Underlining
- Margins
- Avoid graphics and columns

Experience Descriptions

- Action Words
- Quantify
- Outcomes

Describing Your Experience

- *Responsible for several fundraising events*
- *Organized and managed three annual campus and community fundraisers, totaling \$5,000 in donations for charity*

- *Assisted with marketing for fundraising event*
- *Developed marketing campaign composed of social media and print advertisements for silent auctions and walk-a-thons in collaboration with executive board*

Action Verb	+	What you did	+	Why/How/Results
Organized and managed		three annual campus and community fundraisers		totaling \$5,000 in donations for charity

Engineering Physics Examples

- SKILLS

Programming: Python (Pytorch ecosystem, Numpy, Pandas, Scipy (ODE solver), Scikit-learn), MATLAB, Bash Scripting

Software: Linux Command Line, Fusion 360, Git, Slurm, (CAD/FEA), EPOCH (PIC), Multisim

Data, ML, and AI: Genetic Algorithms, Bayesian Optimization (Gaussian Processes), Reinforcement Learning, Neural Networks, Ensemble Methods (Random Forests, Gradient Boosting)

Hardware and Lab: CNC & Manual Machining, 3D Printing, Laser Cutting; Analog & Digital Electronics, Soldering, Oscilloscope, Embedded Python; Electrochemistry (Voltammetry) & Material Synthesis

Engineering Physics Examples

PROJECT EXPERIENCE

High Powered Rocket deploying reinforcement learning drone for rapid Martian mapping

- Lead a senior honors thesis to design, build, and launch a high-powered rocket (640 Ns impulse) with a trajectory control system using telescoping fins
- Developed an autonomous drone payload, deployed at apogee, to perform rapid surface mapping, addressing the capability gap between slow rovers and low-resolution satellites
- Implemented a reinforcement learning model on an onboard Jetson Nano for real-time navigation and safe landing

Engineering Physics Examples

RELATED EXPERIENCE

Computational Analysis Intern, Corvid Technology, Huntsville, AL, May 20XX – Aug 20XX

- Developed a sophisticated Genetic Algorithm with an island model to optimize parameters for a highly noisy, non-differentiable, loss function of a radar tracking software
- Engineered the algorithm to partition the search space, enabling parallel population evolution to locate multiple optima and avoid convergence in local minima, ensuring robust system-wide optimization

Engineering Physics Examples

RESEARCH EXPERIENCE

Undergraduate Researcher, MakerSpaces at William & Mary

Williamsburg, VA, May 20XX - Present

- Operated independently and coordinate with a small team of undergraduates to develop/retrofit a 3D printer to print Piezoelectric PVDF with the goal of printing complex geometries of Piezo-active PVDF
- Designed unique components to allow for operation under high voltage such as ceramic Olsson blocks
- Funded by \$3,000 James Monroe Scholar Upperclassman Grant for summer months

Engineering Physics Examples

RELATED EXPERIENCE

President, William & Mary Rocket Club, Williamsburg VA, February 2024 – Present

- Lead weekly club meetings and direct semester-long projects, including the flagship 'Mega Rocket' initiative
- Spearheaded the 'Mega Rocket' project: developing a high-power rocket with live telemetry, video data transmission to a ground control station, and articulating fins for real-time course correction to maximize apogee.
- Engineered a new cowling and fin design for a standard low-power kit, achieving a 16% simulated apogee improvement (RocketSim); physical test validation pending.
- Designed, sourced, prototyped, and tested custom flight control circuitry for the 'Mega Rocket,' validating functionality using open-source Ardupilot drone firmware.
- Foster member growth by delegating 'Mega Rocket' tasks that align individual strengths with new learning concepts, ensuring project progression and skill development

Let's practice

Create a bullet point description for one of your experiences, resume style.

Networking



Why Network?

- Networking is a key factor in obtaining a **high percentage of all jobs**
- Networking builds and develops communication skills
- Networking should be a part of your lifelong career management



The value of networking has not changed!

Networking is not...

- Asking for a job
- A quick process
- A one-way process
- For outgoing people only
- Rocket science

Networking is...

- Building relationships
- Planned or unplanned
- A two-way process
- For all types of personalities
- A mindset

Networking Opportunities – Intentional/Structured

- Employer Information Sessions/Career Fairs
- Meet Ups and Alumni Panels
- “Coffee Chats”/Quick Conversations
- Conferences
- Joining a group with a mutual interest
- LinkedIn, One Network, and other online channels

Networking Opportunities – Not as Intentional

- Informational Interviews
- Volunteering
- Chance Encounters
 - Paul Simon T-shirt and an airport delay

Build Your Network

- Build your Network – who do you already know?
 - Family members
 - Mentors
 - Friends
 - Professors
 - Campus staff
 - Alumni
 - Neighbors
 - Service providers



& Stay Connected

- Settings
- Connect LinkedIn

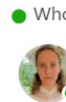
- Home
- Feed**
- Directory
- Career Connections
- Jobs
- Photos
- Communities
- Events
- Business Directory
- Resources
- Info & Support

My Feed

Search Feed



Don: What do you need? What can you offer? Share an insight here.



Who's online • See who's been online recently

SEE ALL LIVE MEMBERS



Alison Moye She, Her, Hers
Chief Operating Officer, 11:59
March 19, 2026, 11:54 AM

Hello Tribe,

This Women's History Month, **11:59** is celebrating the women who are not just participating in the AI and transformation conversation — they're leading it.

We're hosting a **free three-part virtual webinar** ...

**SUPPORT THE
W&M ALUMNI
CAREER
DEVELOPMENT FUND**



Help power programs like Professional Development Week, Career Check Ups, and W&M@Work—providing lifelong career support for the W&M alumni community.

FUEL ALUMNI CAREERS

**W&M
WEBINARS**



William & Mary One Network

- Exclusive to the William & Mary Community
- Can see how people are willing to help
- Can use it individually or as a student organization representative
- Can directly message everyone
- Robust filtering

LinkedIn



- LinkedIn has become a necessary part of almost ANY job or internship search
- Employers will search on LinkedIn for candidates
- LinkedIn allows you to expand your professional network

LinkedIn can...



- Be a career exploration tool
- Help you learn about graduate & professional school
- Help you proactively ask for advice
- Be a “narrative” platform
- Visualize your experiences

Professional Headshot Booth

- Located in career center atrium
- Self-serve, open M-F, 8:00am – 4:45pm
- High-quality photos emailed to you!
- Use for LinkedIn profile and other professional materials



Engineering Physics Alumni Panel

- November 2, 4:00 p.m.
- Cohen Career Center

Next Steps

- *What is one thing you plan to do in the next week to reflect or act on the information you've learned today?*

TribeCareers.wm.edu

Next Steps



*Make an appointment in Tribe Careers
or stop by Quick Advising*



*Subscribe to Industry Newsletters in
TribeCareers
My Account – Industry Newsletters*



*Sign up for Blackboard industry and affinity
pages (including internships)
Search the Course Catalog in Blackboard*



Hours: Monday - Friday, 8:00 a.m. – 5:00 p.m.

Quick Advising Hours: Monday - Thursday, 2:00 - 4:00 p.m.

WILLIAM & MARY

CAREER DEVELOPMENT &
PROFESSIONAL ENGAGEMENT